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EDUCATION

Yale School of the Environment, New Haven, CT
PhD in Environmental Economics, Expected 2026
Committee: Eli P. Fenichel (Chair), Kenneth Gillingham and Arianna Salazar-Miranda
Dissertation Title: Essays in Natural Resource Economics

Yale School of the Environment, New Haven, CT
Master of Environmental Science, 2021
Thesis Title: People Use Park with People: Usage Change during the COVID-19 Pandemic

Montana State University, Bozeman, MT
Bachelor of Science in Economics, Summa Cum Laude, 2018
Minor in Computer Science

JOB MARKET PAPER

Do Hot Deals Beat the Heat? Does Green Space Keep Bringing in the Green? The Effects of Heat and Green Space on City Storefronts: Climate change is increasing the frequency and intensity of extreme heat events, particularly in cities. This paper estimates the causal effect of temperature on daily revenue using over 15,000 consumer-facing storefronts in the 49 largest U.S. cities between 2019 and 2023. Above 35 °C (95 °F), revenue falls steeply, averaging 9 percent lower on days above 37.5 °C (99.5 °F) days than on 20 °C (68 °F) days. Consumption smoothing across days mitigates some damage from an extreme heat event, but a 1.3 percent revenue drop persists for two weeks following a hot day. Therefore, I estimate how effective urban green space is as a climate adaptation strategy that can prevent revenue losses caused by extreme heat ex ante. I find that a one percent increase in tree cover surrounding a storefront increases revenue by 0.94 percent on hot days. A 10 percent increase in surrounding tree cover eliminates revenue losses at 37.5 °C and reduces them by half at 40 °C, relative to areas without canopy. Storefronts located in the Southwest and South have the most to gain from urban tree cover. On average, businesses in this region can cover the cost of moving to a 10 percent tree cover scenario in four years with the avoided revenue losses. This length of time shortens under projected climate scenarios. These results suggest that green infrastructure can improve firm resilience to heat, providing evidence of a private incentive to finance urban green space that could simultaneously provide a positive externality.

WORK IN PROGRESS

1. The National Value of Local Outdoor Recreation over the Past 20 Years (Dissertation Chapter)

2. Measuring the Value of Outdoor Recreation for National Environmental-Economic Statistics (with Eli P. Fenichel and Jorge Forero Fajardo)

PUBLICATIONS

Creel A. M. (2025). Muller et al., eds., Measuring and Accounting for Environmental Public Goods, Chap. 7 Comment. *National Bureau of Economic Research Book, University of Chicago Press.*

Creel, S., Creel, N. M., Creel, A. M., & Creel, B. M. (2016). Hunting on a hot day: effects of temperature on interactions between African wild dogs and their prey. *Ecology*, 97(11), 2910-2916.

RESEARCH FELLOWSHIPS

2024 Graduate Research Fellow, **Property and Environment Research Center** (PERC), Bozeman, MT.

2022 Graduate Research Fellow to **Eli P. Fenichel**, Yale School of the Environment and **Jude Bayham**, Colorado State University. *Analyzed mobility data on the daily dairies of individuals to identify close contacts of meat packer employees during the COVID-19 pandemic.*

2021 Research Associate to **Margaret Walls**, Resources for the Future, Washington D.C. *Examined distribution of Land Water Conservation Funds across sub-populations.*

2019 Graduate Research Associate to **Kerwin Charles**, Yale School of Management, New Haven, CT. *Collected primary source data to analyze the effect of closing Black hospitals during desegregation.*

CONFERENCE AND SEMINAR PRESENTATIONS

2025 CU Environmental and Resource Economics Workshop, Vail, CO
Camp Resources, Asheville, NC
AERE Summer Conference, Santa Ana Pueblo, NM
Northeast Workshop on Energy Policy and Environmental Economics, New Haven, CT
W5133 Economic Valuation and Management of Natural Resources, Burlington, VT

2024 Invited Speaker, Western Community and Connections Seminar Series, Yale School of the Environment
PERC Summer Seminar, Bozeman, MT

2023 Camp Resources, Ashville, NC
OSWEET Series: April 28

2021 OSWEET Series: August 27
PERC Seminar Seminar, Bozeman, MT
AERE Summer Conference (Virtual due to COVID-19 Pandemic)

SELECT AWARDS AND HONORS

2025 Institute for Humane Studies Expense Support Grant

2023 Yale Environmental Data Science Initiative Small Grant
Teaching Fellow Award for Excellence, Yale School of the Environment
Yale Institute for Biospheric Studies Small Grant
Spring Alumni Feature, Montana State University Economics Departments' Magazine. Link.

2021 Class of '21 Student Highlight, Yale School of the Environment. Link.

2020 Alumni Association Board Scholarship, Yale School of the Environment
Schiff Fund for Wildlife, Habitat and Biodiversity Research, Yale School of the Environment

2019 Merit Scholar, Yale School of the Environment
Runner-Up for Best Digital Reporting, International Association of Fire Fighters

TEACHING EXPERIENCE

2025 **Instructor, Certificate in Environmental Data Science, Yale School of the Environment**
Prepared and presented 8 lectures for the data foundations unit, February 2025.
Link to Lecture Notes

Instructor, Intro to Programming Workshop, Yale School of the Environment
Prepared and presented 6 lectures over 3 days, January 2024 and 2025
4.7/5 average review from a total of 28 students.
Link to Lecture Notes

Instructor, Intro to Excel for Environmental Science, Yale School of the Environment
Prepared and presented a two-hour workshop on using Excel for environmental problems, Summer 2024 and 2025

Graduate Teaching Fellow, ENV 762, Yale School of the Environment
Applied Math for Environmental Scientists, Fall 2023, 2024 and 2025
4.8/5 average review from 11 students in 2023. 4.6/5 from 5 students in 2024.
Link to Prepared Notes

Lead Teaching Fellow for Community Orientation (aka MODs), Yale School of the Environment
Co-developed teaching material with faculty member for three-day course in environmental leadership, Summer 2023, 2024, 2025
4.83/5 instructor average review from 53 students in 2024 (2023 and '25 surveys not available)

2024 **Graduate Teaching Fellow, ENV 617, Yale School of the Environment**
Real World Environmental Data Science, Spring 2024
4.6/5 average review from 17 students

2023 **Graduate Teaching Fellow, ENV 795, Yale School of the Environment**
Nature as Capital, Spring 2023
5/5 average review from 16 students (Received Teaching Fellow Award for Excellence)
Link to Prepared Notes

2020 **Graduate Teaching Fellow, ECNS 330, Yale Economics Department**
Natural Resource Economics, Fall 2020
4.7/5 average review from 33 students

2017 **Undergraduate Peer Leader, ECNS 101, MSU Economics Department**
Economic Way of Thinking, Spring, Fall 2017

Undergraduate Teaching Assistant, CSCI 111, MSU Computer Science Department
Introduction to Programming with Java, Spring 2017

2016 **Undergraduate Teaching Assistant, ECNS 204, MSU Economics Department**
Microeconomics, Spring, Summer, Fall 2015; Spring 2016

PROFESSIONAL ACTIVITIES AND ACADEMIC SERVICE

Referee Journal of Environmental Economics and Management

2025 - Present **Founder and Board member of YEEHAW** (Yale Environmental Economics Has a Workshop)
Founded with the mission to build a strong alumni network and foster research collaboration across cohorts of graduates from the Yale School of the Environment

2024 – 2025 **Student Representative on Faculty Search Committee**, Yale School of the Environment

Participated with 3 faculty members to hire an economist and coordinated student engagement

2023 – 2024 **Chair of Economics Student Interest Group**, Yale School of the Environment
Recruited and hosted eleven student seminars and seven group dinners

2023 – 2024 **Supervisor to two Research Assistants**, Yale School of the Environment
Managed an undergraduate and masters student on two separate research projects related to local recreation.

2023 – 2024 **Lead Student Coordinator for Western Community and Connect Center**, Yale School of the Environment
Hosted 18 events for 600 Yale affiliates, created network opportunities with 10 faculty members/practitioners

2019 – 2021 **Secretary of BIOMES**, Yale School of the Environment
Invited and hosted speakers to Yale School of the Environment's flagship seminar series

2020 – 2021 **Chair of the Student Affairs Committee**, Yale School of the Environment
Revised bylaws to define and fund affinity groups, required leadership training for student organizations funded by SAC

SELECT NONPROFIT EXPERIENCE

2018 – 2025 **Backcountry Squatters**, Bozeman, MT
Founder, President of Board
Founded with the mission to grow college-aged women and non-binary folks' participation, leadership, and representation in the outdoor industry and community. Grown to have chapters at 13 universities, and over 2000 members or alumni during my presidency.

2020 **Madera/Chowchilla Resource Conservation District**, San Joaquin Valley, CA
Research Fellow
Contributed to the RCDs' strategic plan aimed at preserving working landowners' livelihoods while also addressing environmental stressors, primarily groundwater depletion.

2017 – 2019 **The Montana Engagement Partnership: Prairie Populist**, Bozeman, MT
Advocacy Journalist
Published one article per week highlighting the positive solutions coming out of Montana's rural communities.

RELEVANT PODCASTS, LETTERS OF OPINIONS, AND TESTIMONY

December, 2024 "Andie — Letting Go of the Familiar to Make Room for What is Next"
What The Hell Do I Do Now?
Link to Podcast

October, 2024 "Promoting the Theory and Practice of Environmental Conservation with the Future Dr. Andie Creel"
Heartwood Podcast
Link to Podcast

February, 2020 "Let's Remain Optimistic in Search for Our Next President"
Letter to the Editor of the Bozeman Daily Chronicle

April, 2019 Testified at Montana State Senate Fish and Game Committee on SJ-7
Study resolution on the value and costs of wolves and grizzlies in Montana
Link to video of testimony

	Covered by the Billings Gazette
May, 2018	"Open Space Reminds us to be Stewards of the Land" *Letter to the Editor of the Bozeman Daily Chronicle*
March, 2017	Testified at Montana State House Natural Resources Committee on HJ-7 Resolution calling on Congress to release wilderness study areas across Montana Link to published testimony as a blog post
January, 2017	"Return the Favor and Protect Public Lands" *Letter to the Editor of the Bozeman Daily Chronicle*
October, 2016	"We Must Ask Ourselves, Where Will the Children Go?" *Opinion to the Editor of the Bozeman Daily Chronicle*

REFERENCES

- **Prof. Eli P. Fenichel**
  Knobloch Family Professor of Natural Resource Economics
  Yale School of the Environment
  eli.fenichel@yale.edu
  (203) 432-5114

- **Prof. Kenneth Gillingham**
  Senior Associate Dean of Academic Affairs; Grinstein Class of 1954 Professor of Environmental and Energy Economics
  Yale School of the Environment
  kenneth.gillingham@yale.edu
  (203) 436-5465

- **Prof. Arianna Salazar-Miranda**
  Assistant Professor of Urban Planning and Data Science
  Yale School of the Environment
  arianna.salazar-miranda@yale.edu
  (203) 436-4569